

Feasibility Memo - Outline

1. Verdict

"Proceed to a baseline triage model, with caveats" - the data has a real target label, complete vitals, and a documented pipeline.

2. Dataset summary

- 55,121 ED encounters, ~225 features (25 structured + ~200 sparse chief-complaint flags)
- ESI (target) has 5 levels; heavily mid-acuity- 49% ESI 3, 32.5% ESI 2, few ESI 1 (0.1%) or ESI 5 (2.2%)
- Adults only (18–107 years), mean age ~55
- Temperature recorded in °F, not °C

3. Top 3 concerns

- **Missingness:** none found in structured columns- surprisingly clean, likely pre-imputed at source (worth stating explicitly, since real-world data usually isn't this tidy)
- **Outcome leakage:** disposition and previousdispo are only known after the visit-excluded as inputs
- **Representativeness:** US academic-hospital sample; case-mix and demographics may not transfer to a Caribbean ED

4. Top 3 reasons to proceed

- Real triage label (esi) present for essentially all rows
- Vitals are complete and show expected separation by ESI (see boxplot grid. e.g. respiratory rate and SpO₂ show the clearest acuity signal in the correlation heatmap: triage_vital_o2 correlates 0.18 with ESI, triage_vital_rr -0.10)
- Documented, reproducible cleaning/EDA pipeline so far

5. Caveats

- Fahrenheit units- thresholds differ from °C
- ~200 chief-complaint flags are sparse; many near-constant (low signal individually)
- Correlations shown are association only, not causal, and not model performance
- External validity for Caribbean deployment is untested

Preliminary feature shortlist (top 10, hypothesis only)

Based on your correlation heatmap and clinical intuition- rank by strength of association with ESI, excluding leakage columns (disposition, previousdispo):

1. triage_vital_o2 (r = 0.18 - strongest single vital correlation)
2. triage_vital_rr
3. triage_vital_hr
4. triage_glucose
5. cc_chestpain
6. cc_shortnessofbreath

7. age
8. cc_abdominalpain
9. triage_vital_temp
10. arrivalmode

(Present as "worth testing in Week 6," not "proven important" — correlation $\neq$ causation.)